

March Madness Variable Documentation

This dataset was filtered to March Madness tournaments between 2008 and 2025 (excluding 2020 since it was canceled). All data was pulled prior to the start of that respective year's tournament start date and is strategically weighted for the 1st half, 2nd half/home, and 2nd half/away parts of the season.

The HoopR package was used to create the dataframe.

General team information

- season = year of the March Madness tournament
- team_id = unique ID of school
- team_name = name of school

Standard basketball scoring metrics

- fg_pct = mean field goal percentage
- three_pt_pct = mean 3-point percentage
- ft_pct = mean free throw percentage
- reb = mean total rebounds per game
- oreb = mean offensive rebounds per game
- dreb = mean defensive rebounds per game
- ast = mean assists per game
- stl = mean steals per game
- blk = mean blocks per game
- to = mean turnovers per game
- pers_fouls = mean personal fouls per game
- ppg = mean points per game
- opp_ppg = mean opponent points per game
- opp_fg_pct = mean opponent field goal percentage
- opp_three_pt_pct = mean opponent 3-point percentage
- efg_pct = mean effective field-goal percentage
 - **Formula:** $(\text{field_goals_made} + (0.5 * \text{three_pt_field_goals_made})) / \text{field_goals_attempted}$
- mov = mean margin of victory per game
- win_pct = win-loss percentage

Advanced basketball metrics

- pace = mean possessions per game

- **Formula:** $\text{field_goals_attempted} - \text{offensive rebounds} + \text{turnovers} + (0.455 * \text{free_throws_attempted})$
 - Reference: Ken Pomeroy <https://kenpom.com/blog/the-possession/>
- unadj_off_eff = unadjusted offensive efficiency
 - **Formula:** points scored / possession
- unadj_def_eff = unadjusted defensive efficiency
 - **Formula:** points allowed / possession
- extraScoreChances = mean extra scoring chances per game
 - **Formula:** offensive rebounds + opponent turnovers - opponent offensive rebounds – turnovers
- oppTO_teamTO_ratio = opponent turnovers:team turnovers ratio
- ast_TO_ratio = assists:team turnovers ratio
- starters_shooting_threes_pct = percentage of starters shooting above the median three-point attempts per game
- three_pt_shooters_at_35_pct_pct = percentage of starters shooting above the median three-point attempts per game AND shooting $\geq 35\%$ from the arc
- close_pct = mean games decided by ≤ 5 points
- blowout_pct = mean games decided by ≥ 20 points
- adj_bayes_win_rate_close = mean close game win percentage
 - Adjusts the win % based on # of games played. If 0, then the global rate
 - Note: I used Bayesian shrinkage formula to calculate close game win %
 - $\text{Adj_win_rate} = (\text{wins} + \text{global_rate} * \text{prior_strength}) / (\text{games} + \text{prior_strength})$
- foul_outs = total number of times players fouled out
- pct_guards = percentage of starters who are guards
- kill_shot_count = per Evan Miyakawa, the kill-shot count is a count of 10-0 runs a team goes on
- healthy_rate = how many total season minutes are accounted for by players who did not play in every game. Not a guarantee that they are injured, but they did miss time.
 - This is flawed by late season injuries, where a player didn't miss many games but had a lot of minutes AND misses the tournament.
- pyth_exp = per Bill James, the expected win-loss percentage given the total points scored and total points allowed on a season
 - I chose an exponent of 11
- luck = the difference between the expected and actual win-loss percentage
- BARTHAG – Bart Torvik variable that factors adjusted OE and DE

- WAB – Bart Torvik wins above bubble (how many more hypothetical wins a team would have vs. a bubble team in the tournament)
- Adjusted OE – adjusted offensive efficiency. Factors in strength of schedule and opponents
- Adjusted DE - adjusted defensive efficiency. Factors in strength of schedule and opponents

Additional team information

- starters = players with respectable rotation playing time
 - **Second Half of Season:** ≥ 14 minutes per game (HOME OR AWAY), ≥ 3 games played, and ≥ 45 minutes played total (HOME AND AWAY)
 - **First Half of Season:** ≥ 12 minutes per game, ≥ 4 games played, and ≥ 60 minutes played total
- Conference = what conference is the team a part of?
- power_conf = binary flag for Big12, ACC, and SEC team
- power_conf_extra = binary flag for Big12, ACC, SEC, + Big Ten, Big East team
- conf_perf = performance in school's conference tournament
 - Specifies win/loss in the Final, Semifinal, Quarterfinal, Early Round – or if they did not play in the conference tournament
- coach_career_adj_win_pct = current coach career win percentage, adjusted for years of experience. Less experience gets higher weight for global win percentage, while more experience has less weight on a global win percentage.
- coach_career_log_games_total = natural log of total career games coached
- coach_career_adj_march_eff = coach performance in March Madness in career, adjusted for years of experience. Performance is multiplied by the following weights:
 - 1.0 = tourney appearance
 - 2.0 = Sweet 16 appearance
 - 3.5 = Final 4 appearance
 - 5.0 = Won NCAA championship
- coach_curr_school_march_eff = coach performance in March Madness at current school, adjusted for years of experience. Same multipliers as above.
- avg_team_exp = average years of experience for rotation players
 - Played ≥ 150 minutes and played ≥ 7.5 minutes per game
- smdi = starters-minutes discrepancy index. Shows the % of official starting 5 that also led the team in minutes per game
 - 0 = same 5 players start and play the most minutes
 - 0.4 = 2 of the top players in minutes played comes off the bench

- 1 = lineup starters $\neq$ rotation core
- womens_team_tourney_flag = did the women's team also make the tournament?
- **(TARGET)** tourney_summary = performance in March Madness tournament
- tourney_perf = round achieved in March Madness tournament